

A smoothing of the Grünbaum Stanley–Reisner threefold

Standalone proof and exact computational certificate

Abstract

We prove that the Stanley–Reisner threefold of the Grünbaum–Sreedharan eight-vertex triangulated 3-sphere is the special fibre of a flat projective family over $\mathbf{Q}[[s]]$ with smooth geometric generic fibre. The proof has three logically separate parts. First, an intrinsic André–Quillen bracket calculation and the curved Bianchi identity construct an all-orders formal deformation. Second, 280 exact singular-incidence calculations show that its rational two-jet already forces generic smoothness, independently of all higher coefficients. Third, Grothendieck existence algebraizes the formal closed subscheme, and a torsion argument proves flatness. All L_∞ -algebras below use the ordinary unshifted cohomological convention: ℓ_n has degree $2 - n$. Every computer-assisted assertion is identified by its exact input, algorithm, output, and mathematical consequence.

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1 The special fibre and the computational contract

Put $k = \mathbf{Q}$, $S = k[x_1, \dots, x_8]$, and let Δ have the following twenty tetrahedral facets:

1234, 1237, 1267, 1347, 1567, 2345, 2367, 3467, 3456, 4567,
2358, 2368, 3568, 1268, 1568, 1248, 2458, 1478, 1578, 4578.

This is the Grünbaum–Sreedharan triangulation in the labelling fixed in [1]. Its minimal nonfaces are the sixteen triples occurring in

$$\begin{aligned}
 I = (& x_6x_7x_8, x_4x_6x_8, x_3x_7x_8, x_3x_5x_7, x_3x_4x_8, x_2x_7x_8, \\
 & x_2x_5x_7, x_2x_5x_6, x_2x_4x_7, x_2x_4x_6, x_1x_4x_6, x_1x_4x_5, \\
 & x_1x_3x_8, x_1x_3x_6, x_1x_3x_5, x_1x_2x_5) \subset S.
 \end{aligned}
 \tag{1.1}$$

Denote the ordered cubic generators displayed in (1.1) by $m_0, \dots, m_{15}$. Write $A = S/I$ and $X_0 = \text{Proj } A \subset \mathbf{P}_k^7$.

Exact computation 1.1 (special-fibre combinatorics). The input of `code/verify_combinatorics.py` is exactly the twenty lines of `data/grunbaum_facets.txt`. The script enumerates all faces and all inclusion-minimal nonfaces. For every face σ , including the empty face, it constructs the simplicial boundary matrices of $\text{lk}_\Delta(\sigma)$ and row-reduces them over $\mathbf{Q}$ using `Fraction` arithmetic. Thus no floating-point calculation occurs.

The output is: 8 vertices, 20 facets, 16 minimal nonfaces (precisely the triples in (1.1)), 97 faces in total, pure dimension 3, every ridge in exactly two facets, connected facet-dual graph, and

$$\tilde{b}(\Delta; \mathbf{Q}) = (0, 0, 0, 1), \quad \tilde{H}_i(\text{lk}_\Delta \sigma; \mathbf{Q}) = 0 \quad (i < \dim \text{lk}_\Delta \sigma)$$

for every σ . By Reisner's criterion [5], A is Cohen–Macaulay of Krull dimension 4. Consequently X_0 is a pure Cohen–Macaulay projective threefold. The certificate is `verification/combinatorics_QQ.txt`.

We use intrinsic André–Quillen cohomology. Let $P \rightarrow A$ be a semi-free commutative Tate differential graded k -algebra resolution, homologically graded with differential of degree -1 . A derivation of cohomological degree p lowers homological degree by p . Set

$$C^p = \text{Der}_k^p(P, A)_0, \quad T_0^p(A) = H^p(C^\bullet), \quad (1.2)$$

where the subscript denotes internal degree zero. Formula (1.2), rather than a naive higher Ext group of I/I^2 , is the definition used below.

Remark 1.2 (an intentionally absent claim). We do *not* claim that $\dim T_0^3 = 24$. A computation of $\text{Ext}_A^2(I/I^2, A)_0$ gives 24, but for a non-lci quotient that group is not automatically intrinsic T^3 . The proof needs only the twelve independent intrinsic T^3 classes established in Proposition 2.3.

2 Construction of the all-orders formal deformation

2.1 The intrinsic tangent Lie algebra and the unshifted convention

The dg Lie algebra $\mathfrak{g} = \text{Der}_k(P, P)_0$, with differential $[d, -]$ and graded commutator, controls graded deformations of A ; this is the tangent Lie algebra construction for a cofibrant resolution [2, 3]. Since $\Omega_{P/k}$ is cofibrant as a dg P -module, applying $\text{Hom}_P(\Omega_{P/k}, -)$ to the quasi-isomorphism $P \rightarrow A$ gives a quasi-isomorphism

$$\text{Der}_k(P, P)_0 \longrightarrow \text{Der}_k(P, A)_0. \quad (2.1)$$

Thus the bracket induced on the cohomology of (1.2) is the intrinsic André–Quillen tangent-Lie bracket. It is not Yoneda multiplication on a conormal Ext group.

Transfer $\mathfrak{g}$ to its cohomology H . We use the ordinary cohomological L_∞ convention throughout:

$$\ell_n : \bigwedge^n H \longrightarrow H, \quad |\ell_n| = 2 - n.$$

The model is minimal, so $\ell_1 = 0$. For $t \in H^1 \hat{\otimes} \mathfrak{m}$ define

$$\kappa(t) = \sum_{n \geq 2} \frac{1}{n!} \ell_n(t, \dots, t) \in H^2 \hat{\otimes} \mathfrak{m}^2, \quad (2.2)$$

and, for $w \in H^2$,

$$d_t(w) = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\underbrace{t, \dots, t}_n, w) \in H^3. \quad (2.3)$$

The L_∞ identities, applied to the curvature (2.2), give the curved Bianchi identity

$$d_t \kappa(t) = 0. \quad (2.4)$$

Notice that (2.2)–(2.4) are unshifted formulas: Maurer–Cartan elements have degree 1, obstructions degree 2, and Bianchi values degree 3.

The completed Maurer–Cartan base is a hull for the graded deformation functor. Make the change of a minimal hull presentation explicit. If the 27 equations in another set of 53 tangent coordinates are

$$G(t) = U(t) \kappa(\phi(t)),$$

where ϕ has invertible linear part ϕ_1 and $U(0)$ is invertible, then

$$B_G(t) = d_{\phi(t)} \circ U(t)^{-1} \quad \text{satisfies} \quad B_G(t)G(t) = 0.$$

Its term linear in t is $\text{ad}_{\phi_1(t)} \circ U(0)^{-1}$. We henceforth use the coordinates produced by the pinned **VersalDeformations** computation. The constant relation-basis transport in (2.9) identifies its 27 obstruction coordinates with the Tate model, while the direct primary-commutator comparison in Proposition 2.3 checks that the linear operator in the preceding display, in exactly these coordinates, is the computed intrinsic bracket (and that the differential of the quadratic equations is -1 times the primary commutator). Thus the Bianchi induction below is invariant under the presentation change, rather than being an identity assumed of package coordinates. From now on we again write κ and d_t for this transported tuple G and operator B_G ; (2.4) denotes the transported identity as well.

2.2 The rational direction and its initial two-jet

In the computed basis of $T_0^1(A)$, take the explicit vector

$$\begin{aligned} y = (& 1, 3, 1, 2, 6, 1, 1, 1, 1, 1, 21, 28, 1, 5, 10, 55, 66, 15, 1, 18, 2, \\ & 35, 42, 1, 4, 10, 1, 25, 12, 30, 5, 33, 44, 1, 6, 3, 14, 7, 1, 9, 1, 6, \\ & 12, 20, 1, 4, 22, 1, 15, 1, 1, 8, 11). \end{aligned} \quad (2.5)$$

No property of how (2.5) was found is used.

Exact computation 2.1 (the degree-three miniversal calculation). The exact inputs of `code/verify_formal_data.m2` are (1.1) and (2.5). Over $\mathbf{Q}$, `CT1(0,F0)` and `CT2(0,F0)` compute the degree-zero tangent and second cotangent-cohomology bases. The command `versalDeformation` is run through order three with `SmartLift=>false` and `DefParam=>t`. Its algorithm solves the deformation equation and records its obstruction coordinates by exact polynomial and module arithmetic.

The asserted output is

$$\dim T_0^1 = 53, \quad \dim T_0^2 = 27, \quad q(y) = 0, \quad \text{rank } Dq_y = 15, \quad \kappa_3(y) = 0, \quad (2.6)$$

where $q = \kappa_2$ in these coordinates and κ_3 is the cubic base term. The 27 quadratic equations are also checked to be a minimal tuple. The same run specializes the miniversal family along $t = sy$

and writes its sixteen first- and second-order cubic corrections to `data/universal_2jet_QQ.txt`; its SHA-256 digest is

$$40e64e61674b6a4e61f1ea6822dc79327bf4ba397285f84ed8738c4c18cd1795. \quad (2.7)$$

Thus $t_2(s) = sy$ has zero curvature through s^3 and gives the displayed family two-jet. The certificate is `verification/formal_data_QQ.txt`.

2.3 The intrinsic bracket calculation

We now explain both what the computer calculates and why it is intrinsic. The materialized part of the semi-free resolution has:

generators	homological degree	internal degree
$e_1, \dots, e_{16}$	1	3
$f_1, \dots, f_{30}$	2	4
$g_1, \dots, g_{136}$	3	5 (16 times), 6 (120 times).

(2.8)

Here $d(e_i)$ is the i th generator of (1.1); the f_j kill a basis of first syzygies, and the g_a kill all cycles at the next stage.

Positive-degree derivations vanish on the polynomial subring S for degree reasons. The degree-two cochain differential is therefore the linear f -part of the 136 elements $d(g_a)$. The source formula used by CT $\hat{2}$,

$$\frac{\mathrm{Hom}_A((\mathrm{syz} F_0)/(\mathrm{Koszul} \text{ syzygies}), A)}{\mathrm{im}(\mathrm{T} \mathrm{syz} F_0)}, \quad (2.9)$$

is exactly $\ker \delta_2 / \mathrm{im} \delta_1 = T_0^2(A)$. The package syzygy basis and the f -basis in (2.8) are related by a 30×30 matrix of constants. The script solves for it, checks its determinant is a unit, transports all 27 classes by its transpose, and checks their closure on all 136 g_a .

Let θ be the degree-one derivation representing y , and let $\eta_1, \dots, \eta_{27}$ represent the transported T_0^2 basis. The script first solves the closure equations for their values on every generator in (2.8), then computes

$$[\theta, \eta_j](g_a) = \theta(\eta_j(g_a)) - \eta_j(\theta(g_a)). \quad (2.10)$$

It finally reduces the 27 columns (2.10) modulo the complete image of $\delta_2 : C_0^2 \rightarrow C_0^3$.

The following extension argument is essential; without it a calculation on a truncated resolution would not prove a statement in T^3 .

Lemma 2.2 (Tate extension). *A degree- p derivation, $p = 1$ or 2 , which is closed on the generators of (2.8), extends without changing those values to a closed derivation of the full Tate resolution P .*

Proof. Proceed through the remaining Tate generators in increasing homological degree. If z is newly adjoined in degree $n \geq 4$, closure prescribes

$$d(\vartheta z) = (-1)^p \vartheta(dz). \quad (2.11)$$

Indeed, closedness $[d, \vartheta] = 0$ is exactly (2.11). Because $d^2 z = 0$ and ϑ is already closed on earlier generators, the right side of (2.11) is a cycle of homological degree $n - p - 1$. This degree is positive. The Tate resolution is acyclic in positive homological degrees, so the cycle is a boundary. A primitive has degree $n - p < n$, hence uses only generators already constructed. The differential preserves internal degree, so the primitive can be chosen in the required internal homogeneous component. This defines $\vartheta(z)$ and completes the induction. The low-degree cases, where positivity alone would not suffice, are precisely the equations solved and asserted by the script on (2.8). $\square$

By Lemma 2.2, θ and each η_j extend globally. Their commutators are therefore globally closed. Consequently the columns (2.10) lie in $\ker \delta_3$, and the natural map

$$T_0^3(A) = \ker \delta_3 / \operatorname{im} \delta_2 \longrightarrow C_0^3 / \operatorname{im} \delta_2 \quad (2.12)$$

is injective on their span. Rank computed in the right side of (2.12) is therefore their actual rank in intrinsic André–Quillen T_0^3 .

Proposition 2.3 (fixed-direction exactness). *For the direction (2.5), the intrinsic bracket satisfies*

$$\operatorname{rank}(\operatorname{ad}_y : T_0^2(A) \rightarrow T_0^3(A)) = 12, \quad \ker \operatorname{ad}_y = \operatorname{im} Dq_y. \quad (2.13)$$

Proof. The exact inputs of `code/verify_aq_bracket.m2` are again (1.1) and (2.5). The algorithm constructs (2.8), performs the basis transport (2.9), solves all closure equations, forms (2.10), and uses exact Gröbner/module arithmetic over $\mathbf{Q}$ to compute its rank in (2.12). Its output is rank 12.

As an independent coordinate check inside the same run, the primary commutator of θ_y with each of the 53 degree-one basis derivations is computed in T_0^2 . Its matrix is exactly $-Dq_y$ in the obstruction coordinates returned by `VersalDeformations`; the minus sign is the differential convention and does not change its image. The script also checks directly that

$$\operatorname{ad}_y Dq_y = 0, \quad \operatorname{rank} Dq_y = 15. \quad (2.14)$$

Thus $\operatorname{im} Dq_y \subseteq \ker \operatorname{ad}_y$. The domain of ad_y has dimension 27, so its kernel has dimension $27 - 12 = 15$, equal to the dimension of $\operatorname{im} Dq_y$. This proves (2.13). The complete output is `verification/aq_bracket_QQ.txt`. $\square$

2.4 Bianchi induction to every order

Proposition 2.4 (all-orders arc). *There is a series*

$$t(s) = sy + v_2 s^2 + v_3 s^3 + \cdots \in T_0^1(A)[[s]], \quad v_2 = 0, \quad (2.15)$$

over $\mathbf{Q}$ whose curvature vanishes identically and whose induced embedded family has the two-jet (2.7).

Proof. By (2.6), $t_2(s) = sy$ has curvature $O(s^4)$. Suppose inductively that a partial series has been chosen with

$$\kappa(t(s)) = r_N s^N + O(s^{N+1}), \quad N \geq 4. \quad (2.16)$$

In the coefficient of s^{N+1} in (2.4), all terms involving a curvature coefficient below N vanish. Since $t(s) = sy + O(s^2)$ and $\ell_1 = 0$, the only remaining term is

$$\ell_2(y, r_N) = \operatorname{ad}_y(r_N) = 0. \quad (2.17)$$

Proposition 2.3 gives $r_N \in \operatorname{im} Dq_y$. Choose $v_{N-1} \in T_0^1(A)$ with

$$Dq_y(v_{N-1}) = -r_N. \quad (2.18)$$

Adding $v_{N-1}s^{N-1}$ to $t(s)$ changes the coefficient of s^N in the quadratic term by $Dq_y(v_{N-1})$. In an m -ary term with $m \geq 3$, one occurrence of the new coefficient has order at least $(N-1) + (m-1) \geq N+1$; two occurrences have still higher order. Hence (2.18) kills precisely the residual in (2.16) without altering lower orders. Induction constructs (2.15), and completeness gives $\kappa(t(s)) = 0$. Every step solves a rational linear system, so all coefficients lie in $\mathbf{Q}$. Pullback of the formal miniversal family produces compatible flat graded deformations over $\mathbf{Q}[s]/(s^n)$ for every n . $\square$

Lemma 2.5 (the formal ideal has sixteen cubic generators). *Let $R = \mathbf{Q}[[s]]$. The formal embedded family of Proposition 2.4 is generated by homogeneous cubics*

$$F_i = m_i + sg_i + s^2h_i + s^3R_i(s, x) \in R[x_1, \dots, x_8] \quad (0 \leq i < 16). \quad (2.19)$$

Their reductions modulo s^3 are the file (2.7).

Proof. Write $\hat{\mathcal{A}} = \varprojlim_n \mathcal{A}_n$ for the compatible flat graded quotients furnished by the formal miniversal family, and let $\hat{\mathcal{I}}$ be the degreewise kernel of $R[x_1, \dots, x_8] \rightarrow \hat{\mathcal{A}}$. For each degree d , the module $(\mathcal{A}_n)_d$ is finite flat, hence free, over the Artinian local ring $R/(s^n)$, with rank independent of n . Its inverse limit is therefore a finite free R -module. It follows from

$$0 \longrightarrow \hat{\mathcal{I}}_d \longrightarrow R[x_1, \dots, x_8]_d \longrightarrow \hat{\mathcal{A}}_d \longrightarrow 0$$

that $\hat{\mathcal{I}}_d$ is finite free as well.

The special degree-three kernel has basis $m_0, \dots, m_{15}$ from (1.1). The sixteen computed equations modulo s^3 lift, by surjectivity from the finite free module $\hat{\mathcal{I}}_3$, to cubics F_i with exactly the two-jet (2.7). Their reductions modulo s form the displayed special basis, so Nakayama's lemma makes them an R -basis of $\hat{\mathcal{I}}_3$. For every $d \geq 3$, the degree- d part of $\hat{\mathcal{I}}/(F_0, \dots, F_{15})$ is a finite R -module whose reduction modulo s is zero, because the special ideal (1.1) is generated by its cubics. Nakayama again makes this quotient zero. For $d < 3$, the same argument applied directly to $\hat{\mathcal{I}}_d$ gives zero. Hence the sixteen cubics generate the full formal ideal. A degree-three polynomial has only finitely many x -monomials, so its s -adic coefficient series make it an honest element of $R[x_1, \dots, x_8]$, proving the claim. $\square$

3 The geometric generic fibre is smooth

Let

$$F_i^{(2)} = m_i + sg_i + s^2h_i \quad (0 \leq i < 16) \quad (3.1)$$

be the cubics in `data/universal_2jet_QQ.txt`. By Lemma 2.5, the family from Proposition 2.4 has equations

$$F_i = F_i^{(2)} + s^3R_i(s, x), \quad (3.2)$$

with $R_i \in \mathbf{Q}[[s]][x_1, \dots, x_8]$ homogeneous of degree three in x .

On the projective chart $x_v = 1$, write $z_1, \dots, z_7$ for the remaining coordinates and let J be the 7×16 affine Jacobian of the equations. The flat family is Cohen–Macaulay of relative dimension three near the special fibre: this follows from Computation 1.1 and the standard flat Cohen–Macaulay fibre criterion (compare [6, Tag 0E7T]). Thus a point of a fibre is singular exactly when $\text{rank } J \leq 3$. Equivalently, $\ker J^\top \subset k^7$ contains a four-plane.

The Grassmannian $\text{Gr}(4, 7)$ has $\binom{7}{4} = 35$ standard affine charts. On the chart indexed by a four-subset $P \subset \{1, \dots, 7\}$, let K_P be the 7×4 matrix whose rows indexed by P form the identity and whose other entries are twelve variables $a_1, \dots, a_{12}$. For a pair (v, P) the singular-incidence ideal is

$$\mathcal{J}_{v,P} = (F_0, \dots, F_{15}, J^\top K_P). \quad (3.3)$$

It has $16 + 16 \cdot 4 = 80$ displayed generators.

Exact computation 3.1 (all 280 incidence memberships). The exact input is (3.1), verified by digest (2.7). For each of the $8 \cdot 35 = 280$ pairs, `code/verify_incidence_chart.m2` dehomogenizes (3.1), constructs J and K_P , and truncates (3.3) modulo s^3 . Put

$$B = \mathbf{Q}[z_1, \dots, z_7, a_1, \dots, a_{12}].$$

For every generator $q = q_0 + sq_1 + s^2q_2$, multiplication by all elements of $B[s]/(s^3)$ is represented exactly by the three columns

$$(q_0, q_1, q_2)^\top, \quad (0, q_0, q_1)^\top, \quad (0, 0, q_0)^\top. \quad (3.4)$$

The 80 generators therefore give a 3×240 module matrix. Membership of $(0, 0, 1)^\top$ in its column module is exactly the assertion

$$s^2 \in (\mathcal{J}_{v,P}, s^3). \quad (3.5)$$

The driver `code/verify_all_incidence.py` enumerates every pair and uses Macaulay2 exact rational module Gröbner reduction. A zero remainder against a partial Gröbner basis is already a proof of membership: each basis column is obtained by exact module operations from the 240 input columns, and the zero reduction expresses the target in those columns. A nonzero remainder is treated only as inconclusive. Degree six proves 277 pairs; degree eight proves the remaining pairs

$$(v, P) = (2, 7), (6, 12), (8, 16). \quad (3.6)$$

The driver checks that its sorted result contains all 280 pairs. On pair (1, 1) it additionally retains the change matrix and verifies the resulting identity in the original columns (eight coefficients are nonzero). The exact output is `verification/incidence_all_QQ.txt`. The mathematical consequence is (3.5) on every incidence chart.

We spell out why 280 memberships exhaust every possible singular generic point and why the truncation is uniform in (3.2).

Proposition 3.2 (two-jet smoothness). *Every flat projective continuation (3.2) has smooth geometric generic fibre.*

Proof. Suppose a geometric generic singular point exists. The singular-incidence scheme is of finite type over $K = \mathbf{Q}((s))$, so it has a point after some finite extension K'/K . Choose a DVR R' in the integral closure of $\mathbf{Q}[[s]]$ in K' with fraction field K' . Properness of $\mathbf{P}^7$ extends the projective point to $\text{Spec } R'$. Scale its homogeneous coordinates so that they are all integral and at least one, say x_v , is a unit. This puts the section in one of the eight affine projective charts.

At the generic point, singularity means $\dim \ker J^\top \geq 4$; choose a four-dimensional subspace. It is a K' -point of the proper Grassmannian $\text{Gr}(4, 7)$ and hence extends over R' . Its Plücker coordinates may be scaled to be integral with at least one unit. The unit Plücker coordinate selects one of the 35 standard charts, and the twelve entries of K_P on that chart are integral. Thus some one of the 280 pairs contains integral coordinates for both the point and its kernel four-plane. The equations $J^\top K_P = 0$, true over K' , remain true over R' because their entries are integral and R' injects into K' .

Fix this chart. Membership (3.5), lifted from the truncated polynomial ring, is an identity

$$s^2 = \sum_{j=1}^{80} c_j q_j + s^3 c_0 \quad (3.7)$$

with c_j, c_0 polynomial in s, z, a and q_j the truncated incidence generators. For the actual continuation, each generator is $q'_j = q_j + s^3 r_j$: differentiation with respect to an affine coordinate does not lower s -adic order. Substitution in (3.7) gives

$$s^2 = \sum_j c_j q'_j + s^3 \left(c_0 - \sum_j c_j r_j \right). \quad (3.8)$$

All quantities in parentheses evaluate to elements of R' because every projective and Grassmann coordinate is integral and the coefficients of (3.2) lie in $\mathbf{Q}[[s]]$. Along the incidence section all q'_j vanish, so (3.8) becomes

$$s^2 = s^3 c \quad (c \in R'). \quad (3.9)$$

If v is the valuation of R' , then $v(s) > 0$, whereas (3.9) implies $2v(s) = 3v(s) + v(c) \geq 3v(s)$, a contradiction. Hence no geometric generic singular point exists. $\square$

4 Algebraization and flatness

Set $R = \mathbf{Q}[[s]]$ and $R_n = R/(s^n)$. Proposition 2.4 gives cartesian closed subschemes

$$X_n \subset \mathbf{P}_{R_n}^7, \quad X_{n+1} \times_{R_{n+1}} R_n = X_n, \quad (4.1)$$

each flat over R_n , with $X_1 = X_0$. We now justify both effectivity and flatness of the algebraized family.

Proposition 4.1 (effectivity). *The system (4.1) is the system of reductions of a closed subscheme $X \subset \mathbf{P}_R^7$ that is proper over R .*

Proof. The ring R is Noetherian and complete for (s) , and $\mathbf{P}_R^7$ is separated and of finite type (indeed proper) over R . The special member X_1 is projective over $\mathbf{Q}$. Thus every hypothesis of Grothendieck's algebraization theorem for compatible closed subschemes is satisfied; in the formulation [6, Tag 0899], (4.1) algebraizes to a closed immersion $X \hookrightarrow \mathbf{P}_R^7$, and X is proper over R .

For completeness, the theorem applies to the compatible coherent quotients $\mathcal{O}_{\mathbf{P}_{R_n}^7} \twoheadrightarrow \mathcal{O}_{X_n}$. Grothendieck existence algebraizes the quotient and its map; exactness and full faithfulness of completion show that the algebraized map is still surjective, hence its kernel is a coherent ideal sheaf. This is why existence of a coherent sheaf alone is not being mistaken for existence of a closed subscheme.

There is also a concrete description. Lemma 2.5 gives sixteen homogeneous cubics in $R[x_1, \dots, x_8]$. Their Proj has reductions (4.1), and the full-faithfulness part of Grothendieck existence identifies it with the closed subscheme just algebraized. Consequently the equations used in the DVR argument are equations of the algebraized family, not merely unrelated formal functions. $\square$

Proposition 4.2 (flatness). *The morphism $X \rightarrow \operatorname{Spec} R$ is flat.*

Proof. It is enough to prove R -flatness at every stalk $B = \mathcal{O}_{X,x}$. If x is on the generic fibre, s is a unit. Suppose x specializes to the closed fibre and $sb = 0$ in B . For every $n \geq 2$, the reduction $B/s^n B$ is flat over R_n by (4.1). In R_n the annihilator of s is (s^{n-1}) . Tensoring the corresponding kernel sequence with the flat R_n -module $B/s^n B$ gives

$$b \bmod s^n B \in s^{n-1}(B/s^n B).$$

Thus $b \in s^{n-1} B$ for every n . The local ring B is Noetherian and s lies in its Jacobson radical, so Krull intersection gives $\bigcap_n s^n B = 0$ and hence $b = 0$. Therefore B is s -torsion-free. Since R is a DVR, an R -module is flat exactly when it is torsion-free. Thus every stalk is R -flat. $\square$

Theorem 4.3 (smoothing theorem). *The Grünbaum Stanley–Reisner threefold X_0 is smoothable in characteristic zero. More precisely, there is a flat projective morphism*

$$X \longrightarrow \operatorname{Spec} \mathbf{Q}[[s]]$$

whose special fibre is X_0 and whose geometric generic fibre is smooth.

Proof. Proposition 2.4 constructs the compatible formal family; Propositions 4.1 and 4.2 algebraize it and prove flatness; Proposition 3.2 proves smoothness after algebraic closure of the fraction field. The special fibre is (1.1) by construction. $\square$

5 Audit table for computer-assisted lemmas

Claim	Exact input	Exact algorithm and output	Consequence
Special fibre	Twenty facets in data/	Enumerate faces; rational boundary-matrix ranks; 97 Reisner checks	Pure CM threefold; ideal (1.1)
Initial arc data	Ideal (1.1), vector (2.5)	M2 miniversal deformation to degree 3; (2.6), digest (2.7)	Initial zero residual and fixed two-jet
Intrinsic bracket	Same ideal and vector	Three Tate stages; extendable derivations; commutators modulo all δ_2 boundaries; ranks 12 and 15	$\ker \operatorname{ad}_y = \operatorname{im} Dq_y$
Generic smoothness	The regenerated two-jet	280 exact module memberships over $\mathbf{Q}$; 277 at degree 6, 3 at degree 8	DVR contradiction for every higher continuation

All four rows are run by `./verify.sh`. The scripts contain assertions for dimensions, basis changes, closure equations, ranks, chart coverage, semantic output, and hashes. The calculations are indispensable parts of the proof and are not concealed as heuristic evidence.

References

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